RESEARCH-ARTICLE

Remaking Game Art Based on Deep Learning

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Abstract

With the development of computer vision technology, the art level and visual expression of modern electronic games have been greatly improved. The endless stream of electronic games cannot be compared to the first version of games through technological iteration. In the current era where game engine technology and graphics rendering capabilities have been greatly improved, game developers have shifted their focus to the remake of old games, updating the art assets of old games through deep learning algorithms and combining modern game production techniques to reset the games. This article explores the art asset production of remastered games based on deep learning, and reasonably utilizes deep learning to generate new art assets within the scope of avoiding infringement of others' intellectual property rights, greatly shortening the game production cycle.

CCS Concepts

• **Theory of computation** → Theory and algorithms for application domains ~ Machine learning theory ~ Models of learning.

Keywords

game art, Deep learning, Image processing, Computer Vision

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1 Introduction

Game art is one of the core components of electronic game development, responsible for the important role of information exchange between players and gaming devices. As a comprehensive field that highly integrates art and technology, it involves various knowledge from low-level graphics algorithms to high-level engine tools, covering graphics rendering, geometric processing, animation calculation, special effects simulation, etc.The gaming industry has traditionally encountered with high costs and extensive time requirements for producing assets, a crucial component in game development. As artificial intelligence (AI) continues to advance, its applications are becoming increasingly relevant in various domains, including game development[1]. The expressive power of art is constrained by technology, while also reflecting the level of technology at this moment. With the improvement of hardware computing power and optimization of algorithms such as RTX ray tracing and DLSS frame generation, the graphics of electronic games have evolved from 32 * 32 pixel monochrome images to today's high-resolution, high refresh rate, and wide color gamut surreal scenes. More exciting game graphics represent richer visual styles, more realistic atmosphere creation, and a higher quality gaming experience. Excellent game art combined with color psychology, visual guidance, camera language and other techniques can provide players with visual enjoyment while also influencing their emotions and behavior, bringing a better gaming experience.

Due to the flourishing development of data-driven visual content generation technology, computer vision research based on deep learning has had a huge impact on the art industry. The electronic gaming industry has begun to extensively use data-driven visual content generation technology to assist in game art production. In the recent 10 years, the electronic game industry has witnessed a wave of "re plate making" game production. Game companies use modern computer vision technology to "reset" the early popular games with copyrights on the basis of retaining most of the original story, art style, pictures and other contents of the original game, so as to stimulate old players to buy and play again while attracting the attention of new players, so as to ensure the lasting vitality of the game IP. "Reprint" game production mainly focuses on the renovation of game visual effects and fun optimization of game playing methods, which requires a long production cycle, in which the re production of game art assets is the main work content. Taking the game Dead Space as an example "Figure 1", according to the disclosure of the technical director of Motive Studios in the EA official live broadcast, the core development cycle of the remake is about two years. Among them, the picture upgrade part of the game, including the improvement of the number of model faces and the refinement of scene materials, took 8 months, accounting for 19% of the total game production time. Maxon shows in the ZBrush User Behavior Report released in 2022 that the efficiency of carving high detail models for intermediate users is 20-30 faces per minute. Based on this, it will take about 6-8 hours for a 1000 face model to add details to 10000 faces. The optimization of art resources of the "reset version" game based on in-depth learning can not only greatly shorten the production cycle, but also make full use of AI's learning ability. Under the condition of holding the copyright of the original work, reference to the original art resources effectively avoids the intellectual property problems that generative visual art may encounter. Based on in-depth learning, this research conducts

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Early Preparation	Engine Migration	Screen Upgrade	Adjustment	Test
1.5 years	4 months	8 months	5 months	6 months
44%	10%	19%	12%	15%

Figure 1: Time consumption statistics of Dead Space

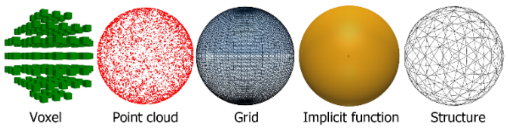

Figure 2: Visual effects in different forms of expression

modern visual content generation of old game art resources for "re plate making" game production, with the purpose of improving the expressiveness of the game screen and continuing the value of the game, providing a more efficient means of game art production for game companies, and also bringing more high-quality game content to game players.

At present, the main applications of deep learning in the field of electronic games are game art content generation and NPC behavior training. Among them, AI has brought tremendous impact to the game art industry, and many game companies have replaced manual digital painting with AI generated images.Tools like Generative Adversarial Networks (GANs) have been particularly successful in generating high-quality textures, characters, and environments[2]. The content is about the optimization of game original art assets using Generative Adversarial Networks (GANs) and Convolutional Neural Networks (CNNs), mainly involving two parts: improving 3D model accuracy and enhancing texture resolution. In the deep learning stage of 3D model generation, the popular structural representations for 3D geometry learning mainly include voxels, point clouds, meshes, implicit functions, and structural representations "Figure 2"[3].

Voxel representation has the characteristics of stable image generation and intuitive representation, but due to current hardware computing limitations, the resolution of 3D voxel grids is usually not high, which may lead to insufficient accuracy of the generated model. Compared to voxel representation, point cloud representation has an accurate shape representation that improves the generation of spatial coordinates, but its representation form is not suitable for current deep learning training. Grid representation, as the current mainstream representation form based on surface elements, has a higher accuracy than point clouds while consuming less memory. Implicit representation of 3D models in the form of functions can continuously improve accuracy through repeated training, but it cannot intuitively represent geometric features. The structural representation represents the positional relationship between strong and simple geometric shapes, which can ensure the stability of the overall shape but is prone to detail loss. In the deep learning stage of model mapping production, the main application scenarios are missing image completion, noise suppression, deblurring, super-resolution, lighting enhancement, etc. The image optimization technology based on convolutional neural networks is still the mainstream technology at present, and has developed from independent models to mixed models that cooperate with various models. Some scholars have proposed a hybrid deep learning model combining CNN and recurrent neural network (RNN) to optimize complex character image details. This method can generate more refined and design-oriented character details through multi-level convolutional feature extraction and temporal information modeling [4].

2 Methods

At present, the most common production process in the video game industry is the Physically Based Rendering (PBR) process. The PBR model production process is a standardized process combining high-precision modeling, physical realistic materials and real-time rendering technology. By baking the surface details of high-precision 3D modeling in the form of maps into normal maps, OA maps, etc., and then assigning the normal maps to the low-precision 3D model from the topology of high-precision maps optimizing the number of faces. The 3D low precision model with normal map and OA map can be recognized and rendered in detail in the engine, which greatly reduces the performance requirements of the equipment while ensuring the image quality.

This study is based on the traditional production process of the electronic gaming industry combined with deep learning image processing models. Using the publicly available game asset library of Unity Asset Store as the dataset, PyTorch is used to build a deep learning framework, and Blender is used as the final effect display platform.The art resource update process is formulated as follows: original game asset extraction → material classification → 2D material preprocessing → noise elimination → super-resolution processing → generation of restored images → 3D model generation → texture generation → style transfer → engine compatibility adjustment → final replacement " Figure 3". The above process mainly applies CNN feature extraction, image restoration technology, GAN model generation, and stylization processing technology to realize the transformation of raw art resources from high-definition concept maps to high-precision mesh models, and finally combined with high-precision textures to form high-definition art resources.

2.1 Preprocessing of original art assets:

Manually extract necessary art assets such as 3D models, model textures, animation sequences, and scene images of the original game, and store the extracted materials in two categories for optimization and reference. The most crucial technology of computer vision technology is image processing, and the most critical point of the image processing technology is image feature point extraction and stereo matching technique[5]. Reference materials typically consist of game concept art sets, unused high-definition concept images, and game screenshots and other art resources. These materials will serve as training datasets for subsequent deep learning models, providing data labels for supervised learning. The optimized materials need to go through the following steps to standardize the unstructured game art assets:

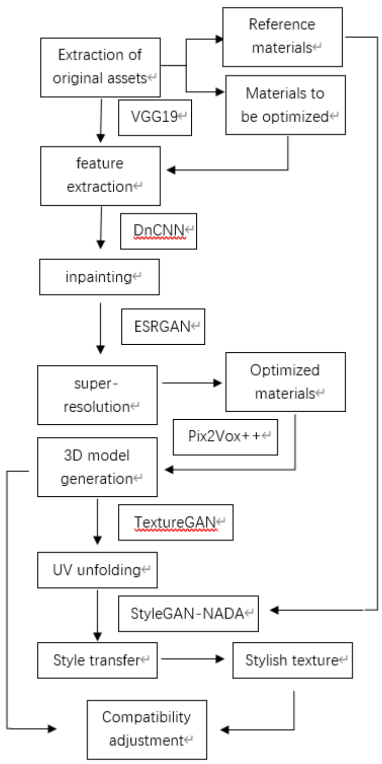

Figure 3: Specific process

First, the Lanczos interpolation algorithm is used for the image to eliminate the jagged effect when scaling the image, and the resolution is unified by using the method of L (x) weighted summation to obtain the interpolation results. The adjacent pixels (| x |<a) are weighted and summed. The larger the a is, the more obvious the details are retained, and the more computation is required. Ensures that the convolution kernel can be accurately recognized in feature extraction. The expression is:

$$L(x) = \begin{cases} \sin c(x) \sin c(\frac{x}{a}) & if 0 < |x| < a \\ 0 & otherwise \end{cases} \quad (1)$$

Among them, a is the range of neighboring pixels considered when controlling the interpolation of filter window size. X is the normalized distance between the target pixel and neighboring pixels, and the relative distance between the texture coordinates and surrounding pixels is processed to determine the weight of each sampling point. C is the truncation coefficient of the interpolation kernel function.

After the unified resolution processing is completed, the image is subjected to color linearization processing. Due to the built-in sRGB decoding function of modern GPUs, texture sampling can be automatically converted through sample2D, and the parts of the gamma curve with a turning threshold below 0.04045 can be linearly processed in the dark area to ensure that there is no color level breakage at the intersection of light and dark. Image quality enhancement:

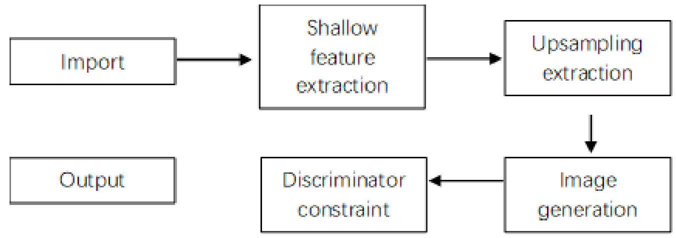

Figure 4: Specific process

In the image quality enhancement stage, the VGG19 deep convolutional network is first used to perform multi-layer convolution and pooling feature extraction on the texture material, analyzing visual features such as image features and color distribution as reference benchmarks for style transfer. Subsequently, the DnCNN deep learning model was used to eliminate block artifacts and noise interference caused by early compression storage in the texture. Calculate the restored image after obtaining the predicted noise through inverse operation.

$$\hat{n} = \arg \frac{\min}{n} ||y - n - x||_2^2 + \lambda R(n) \quad (2)$$

$$\hat{x} = y - \hat{n} \quad (3)$$

Where y is the initial noise image, n is the noise field to be solved, and λR (n) is the regularization term and weight coefficient that constrains the physical properties of the noise. Separating noise $\hat{n}$ from clean image x by $||y - n - x||_2^2$ using residual learning.

For textures with large missing areas, attention based image restoration models such as LaMa are used for content completion and super-resolution reconstruction of the image. In the super-resolution reconstruction stage, the generated results of the adversarial loss constraint generator of the discriminator are used to obtain high-resolution high-definition images"Figure 4".

Compared to traditional SRCNN, it can better handle common sharp edges and repetitive patterns in game textures, effectively improving the resolution of blurry textures while maintaining the authenticity of image details.

2.2 3D model generation:

As the core process of updating game art assets, this step requires generating 3D models based on 2D images. Generative AI has gained prominence for its ability to create complex and diverse game assets in game design by blending human creativity with technical innovation. Tools like Generative Adversarial Networks (GANs) have been particularly successful in generating high-quality textures,characters, and environments[6]. This step uses Pix2Vox++, a neural network architecture designed specifically for multi view reconstruction, combined with NVIDIA Omniverse platform's automated topology repair tool to convert character settings from different angles into editable high-precision models. Firstly, a 3D voxel model is generated by predicting 2D original multi view images through neural networks, and then converted into an editable mesh model. Finally, a high-precision editable model is output

Figure 5: Results show

through topology optimization loss.

$$\begin{aligned} &V = Encoder(I_1, I_2, ..., I_n) \\ &\hat{M} = Decoder(V) \\ &L_{topo} = \lambda_{edge}||E(\hat{M}) - E(M_{gt})||_1 + \lambda_{curv}L_{curvature} \end{aligned} \tag{4}$$

Among them, V is the predicted voxel space, I is the reference view, $\hat{M}$ is the preliminary grid output, M_{gt} is the reference model, L_{topo} is the topology optimization model, λ_{edge} is used to control the proportion of quadrilateral surface, which is usually 0.7, and λ_{curv} is used to ensure the sharpness and detail retention of hard surface, which is usually 0.3.

The final effect is shown in Figure 5. Compared with the original model that uses pre rendered light and shadow maps to assign the model with perspective limitations, the high-precision model generated through deep learning has significantly improved visual aspects such as the number of model faces and texture quality.

2.3 Style transfer:

In the stylization stage, StyleGAN-NADA, a text guided generative model trained on reference materials, is used. Much of the work on GAN architectures has focused on improving the discriminator by, such as using multiple discriminators, multiresolution discrimination, or self-attention. The work on generator side has mostly focused on the exact distribution in the input latent space or shaping the input latent space via Gaussian mixture models, clustering, or encouraging convexity[7]. Compared to traditional CycleGAN, it allows users to accurately control the output effect through natural language commands such as "Pop Art+Ink Style", avoiding randomness in the style transfer process. The StyleGAN-NADA model simultaneously receives supervised learning from the original painting training set to ensure consistent styles across different assets. The core principle is to map natural language instructions to the style space of the generative network through the cross modal alignment capability of the CLIP model.

2.4 Engine compatibility adjustment:

The final stage of engine integration requires close cooperation between manual and technical means, which belongs to the field where AI technology has not yet matured. For example, AI is not yet proficient in checking various parameters of generated assets, focusing on correcting details such as UV seam alignment, material ball attribute settings, and bone weight allocation. At present, in the production of electronic games, the manual processing of game art assets in the engine plays an irreplaceable role. For example, the main character model, key performance scenes, and other generated assets may require separate manual modifications. This systematic processing scheme not only preserves the artistic features of the original work, but also significantly improves the visual quality, providing a scalable technical framework for the remake of classic games and a reference for the integration of deep learning and the game production industry in the future.

3 Summarize

In the traditional game remake process, the art team needs to manually rebuild high-precision models, redraw textures, and adjust lighting, which is time-consuming and labor-intensive. Through AI driven texture enhancement, model optimization, and style transfer techniques, developers can reset the art resources of classic games at a low cost and with high quality, efficiently presenting them to players in next-generation standards. This trend not only reduces the development cost of remastered games, but also significantly shortens the production cycle, injecting new vitality into the gaming industry. However, the GAN architecture suffers in general given its reliance on training data and can have difficulty when working with small datasets[8]. In the future, the application of multimodal generation technology in the generation of complex art assets can be explored, and a more intelligent engine integration toolchain can be developed to further reduce remanufacturing costs. This study provides a scalable technical framework for high-definition classic games, combining artistic value and commercial feasibility.

However, the balance between automation and manual creation, as well as the normalization of style transfer, is an inevitable challenge for the combination of deep learning and game art in this technology.

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